

Syllabus for MA 5104

Statistical Inference and Learning, Fall 2026

INSTRUCTOR INFORMATION

INSTRUCTOR: Rakhi Singh
CLASS MEETING: **(B): Mon @9, Tues @8, Wed @12**, and Fri @11 (3 classes/week)
CLASS LOCATION: KCB 504
OFFICE: KCB 642
E-MAIL: `rakhi@smail.iitm.ac.in`

COURSE INFORMATION

Description. This course develops a unified view of statistical inference and statistical learning, combining classical foundations with modern, computationally driven methods. Students will study estimation, hypothesis testing, and resampling alongside predictive modeling and Bayesian inference, with an emphasis on uncertainty quantification, model assessment, and real-data analysis. It is intended for graduate students who have one probability course and who wish to review the fundamentals before taking additional statistics courses and/or the students who are wish to understand statistics for it being helpful in the industry jobs. By the end of the course, you should expect to

- Develop a rigorous foundation in statistical inference based on probability models.
- Study core ideas of statistical learning, classification, and regularization.
- Develop an ability to apply appropriate statistical methods to summarize and analyze data for some of the more routine experimental settings.
- Integrate theory and computation through applied case studies and projects.
- Make sense of data and be able to report the results in appropriate table or statistical terms for inclusion in your thesis or paper.
- Perform appropriate statistical techniques using R and interpret the results/outputs.

TEXTBOOK(S)

I will primarily be referring to the following two books:

Walpole, Ronald E.. (2012). Probability and Statistics for Engineers and the Scientists 9th ed. (9th ed.). Prentice Hall.

Hastie, Trevor, Tibshirani, Rob and Friedman, Jerome, The Elements of Statistical Learning, 2nd ed., Springer, 2009.

While you are not required to buy these books for the class, it is advisable that you read at least one of the books cover to cover. I will also be occasionally referring to:

Agresti, A., and Kateri, M. (2021). Foundations of statistics for data scientists: with R and Python. Chapman and Hall/CRC.

Casella, George and Berger, Roger L., Statistical Inference, 2nd ed., Duxbury, 2002.

Efron, Bradley, and Trevor Hastie. Computer age statistical inference, student edition: algorithms, evidence, and data science. Vol. 6. Cambridge University Press, 2021.

Some knowledge of the statistical software package R is required. Some parts of the homeworks will typically be in R. There are many online resources where you can learn the basics of R. For example,

- An Introduction to R (<https://cran.r-project.org/doc/manuals/R-intro.pdf>);
- R tutorial by Kelly Black (<https://www.cyclismo.org/tutorial/R/>);
- The undergraduate guide to R by Trevor Martin (<https://biostat.jhsph.edu/~ajaffe/docs/undergradguidetoR.pdf>); and
- a pointer to R bloggers (<https://www.r-bloggers.com/>).

COURSE PLAN

Time permitting, this is the plan for the semester:

Module	Topics
1. Statistical Models and Likelihood	Statistical models; likelihood principle; maximum likelihood estimation; computational statistics.
2. Estimation	Point estimation; properties of estimators; asymptotic theory; bootstrap methods.
3. Inference	Sampling distributions; confidence intervals; hypothesis testing; resampling-based inference.
4. Prediction and Statistical Learning	Regression; classification; loss functions; predictive performance.
5. Statistical Learning Theory and Generalization	Empirical risk minimization; cross-validation; regularization; model selection; ensemble learning.
6. Statistical Thinking in Practice	Reproducible data analysis; statistical communication; case studies; student project presentations.

GRADING

Your grade will be based on your performance on

- quiz 1 (15 percent),
- quiz 2 (15 percent),
- project (15 percent),
- homeworks (10 percent)
- final exam (45 percent),

Homeworks. There will be 5 graded homeworks, you will have approximately a week to submit homeworks. The solutions will need to be handwritten.

Projects: A list of potential projects will be provided, each centered around a publicly available dataset, scientific study, or news article making a quantitative claim. For example, students might investigate claims such as “Does drinking coffee increase life expectancy?” by examining the underlying data, identifying potential sources of bias or confounding, assessing the statistical evidence, and communicating the strengths and limitations of the conclusions. The emphasis is not merely on applying statistical methods, but on asking the right questions, evaluating the validity of statistical claims, and presenting results clearly to both technical and non-technical audiences. Detailed project descriptions will be released after Module 1. Depending on the class size, projects may be completed individually or in groups of at most two students. Assessment will be based primarily on a 5–10 minute presentation during the final week of lectures, together with supporting analysis and code.

Exams: The final end-semester exam will be comprehensive. Quizzes 1 and 2, and examinations will be held in the institute-planned slots. The material covered in quizzes will remain flexible and will be conveyed in due course of time.